



# PREALG: Ratios Proportions and Percents

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## Learning Objectives

- Write and simplify ratios
- Solve proportions using cross-multiplication
- Compute percent of a number
- Solve percent increase/decrease and real-world percent problems

*Simplify each expression completely. Show all steps and circle your final answer.*

## Percent of a number

1. Find 10% of 20.

10% of 20

Answer: \_\_\_\_\_

2. A shirt costs \$100. It is 10% off. How many dollars is the discount?

10% of 100

Answer: \_\_\_\_\_

3. A store is offering a 36% discount on an item priced at \$153. How many dollars is the discount?

36% of 153

Answer: \_\_\_\_\_

4. A fundraiser has raised 37% of its \$263 goal. How many dollars have been raised?

37% of 263

Answer: \_\_\_\_\_

5. Find 25% of 20.

25% of 20

Answer: \_\_\_\_\_

6. A shirt costs \$200. It is 20% off. How many dollars is the discount?

20% of 200

Answer: \_\_\_\_\_



7. A store is offering a 18% discount on an item priced at \$132. How many dollars is the discount?

18% of 132

Answer: \_\_\_\_\_

8. A fundraiser has raised 25% of its \$150 goal. How many dollars have been raised?

25% of 150

Answer: \_\_\_\_\_

9. Find 75% of 60.

75% of 60

Answer: \_\_\_\_\_

10. A shirt costs \$200. It is 50% off. How many dollars is the discount?

50% of 200

Answer: \_\_\_\_\_

11. A store is offering a 40% discount on an item priced at \$51. How many dollars is the discount?

40% of 51

Answer: \_\_\_\_\_

12. A fundraiser has raised 46% of its \$197 goal. How many dollars have been raised?

46% of 197

Answer: \_\_\_\_\_

### Solving proportions

13. Solve the proportion:  $\frac{1}{8} = \frac{1}{x}$ . Find x.

$$\frac{1}{8} = \frac{1}{x}$$

Answer: \_\_\_\_\_

14. A car travels 12 miles on 2 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion  $\frac{2}{12} = \frac{5}{x}$  and solve.

$$\frac{2}{12} = \frac{5}{x}$$

Answer: \_\_\_\_\_



15. If 6 shirts cost \$20, how much do 6 shirts cost? Set up and solve a proportion.

$$\frac{6}{20} = \frac{6}{x}$$

Answer: \_\_\_\_\_

16. Solve the proportion:  $\frac{1}{9} = \frac{6}{x}$ . Find x.

$$\frac{1}{9} = \frac{6}{x}$$

Answer: \_\_\_\_\_

17. A car travels 21 miles on 1 gallons of gas. How many miles can it travel on 6 gallons? Set up a proportion  $\frac{1}{21} = \frac{6}{x}$  and solve.

$$\frac{1}{21} = \frac{6}{x}$$

Answer: \_\_\_\_\_

18. If 6 shirts cost \$16, how much do 7 shirts cost? Set up and solve a proportion.

$$\frac{6}{16} = \frac{7}{x}$$

Answer: \_\_\_\_\_

19. Solve the proportion:  $\frac{1}{6} = \frac{4}{x}$ . Find x.

$$\frac{1}{6} = \frac{4}{x}$$

Answer: \_\_\_\_\_

20. A car travels 13 miles on 1 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion  $\frac{1}{13} = \frac{5}{x}$  and solve.

$$\frac{1}{13} = \frac{5}{x}$$

Answer: \_\_\_\_\_

21. If 5 shirts cost \$40, how much do 12 shirts cost? Set up and solve a proportion.

$$\frac{5}{40} = \frac{12}{x}$$

Answer: \_\_\_\_\_

## Writing and simplifying ratios



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22. Write the ratio 10 to 35 in simplest form.

10 : 35

Answer: \_\_\_\_\_

23. A bag contains 12 red marbles and 12 blue marbles. Write the ratio of red to blue in simplest form.

12 : 12

Answer: \_\_\_\_\_

24. On a team, 12 players are seniors and 28 players are juniors. Write the ratio of seniors to juniors in simplest form.

12 : 28

Answer: \_\_\_\_\_

25. Write the ratio 20 to 15 in simplest form.

20 : 15

Answer: \_\_\_\_\_

26. A bag contains 12 red marbles and 16 blue marbles. Write the ratio of red to blue in simplest form.

12 : 16

Answer: \_\_\_\_\_

27. On a team, 20 players are seniors and 16 players are juniors. Write the ratio of seniors to juniors in simplest form.

20 : 16

Answer: \_\_\_\_\_

28. Write the ratio 3 to 9 in simplest form.

3 : 9

Answer: \_\_\_\_\_

29. A bag contains 4 red marbles and 16 blue marbles. Write the ratio of red to blue in simplest form.

4 : 16

Answer: \_\_\_\_\_



**30.** On a team, 16 players are seniors and 28 players are juniors. Write the ratio of seniors to juniors in simplest form.

**16 : 28**

Answer: \_\_\_\_\_

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# PREALG: Ratios Proportions and Percents

ANSWER KEY & SOLUTIONS

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*Topics: Writing and simplifying ratios, Percent of a number, Solving proportions. All answers verified by independent computation.*

## Solutions

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## Percent of a number

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1. Find 10% of 20.

10% of 20

→ Convert:  $10\% = 10/100$ .

→ Multiply:  $10/100 \times 20 = 2$ .

**Answer:**  $10\% \times 20 \div 100 = 2$

---

2. A shirt costs \$100. It is 10% off. How many dollars is the discount?

10% of 100

→ Discount = 10% of \$100 =  $10/100 \times 100 = \$10$ .

**Answer:**  $10\% \times 100 \div 100 = 10$

---

3. A store is offering a 36% discount on an item priced at \$153. How many dollars is the discount?

36% of 153

→ Discount =  $36\% \times \$153$ .

→ Convert percent:  $36/100 = \text{decimal}$ .

→ Discount =  $36 \times 153 / 100 = \$55$ .

**Answer:**  $36\% \times 153 \div 100 = 55$

---

4. A fundraiser has raised 37% of its \$263 goal. How many dollars have been raised?

37% of 263

→ Amount raised =  $37\% \times \$263$ .

→  $= 37 \times 263 / 100 = \$97$ .

**Answer:**  $37\% \times 263 \div 100 = 97$

---

5. Find 25% of 20.

25% of 20

→ Convert:  $25\% = 25/100$ .

→ Multiply:  $25/100 \times 20 = 5$ .

**Answer:**  $25\% \times 20 \div 100 = 5$

---

6. A shirt costs \$200. It is 20% off. How many dollars is the discount?

20% of 200

→ Discount = 20% of \$200 =  $20/100 \times 200 = \$40$ .

**Answer:**  $20\% \times 200 \div 100 = 40$

---



7. A store is offering a 18% discount on an item priced at \$132. How many dollars is the discount?

18% of 132

→ *Discount* =  $18\% \times \$132$ .

→ *Convert percent:*  $18/100 = \text{decimal}$ .

→ *Discount* =  $18 \times 132 / 100 = \$23$ .

**Answer:**  $18\% \times 132 \div 100 = 23$

---

8. A fundraiser has raised 25% of its \$150 goal. How many dollars have been raised?

25% of 150

→ *Amount raised* =  $25\% \times \$150$ .

→ =  $25 \times 150 / 100 = \$37$ .

**Answer:**  $25\% \times 150 \div 100 = 37$

---

9. Find 75% of 60.

75% of 60

→ *Convert:*  $75\% = 75/100$ .

→ *Multiply:*  $75/100 \times 60 = 45$ .

**Answer:**  $75\% \times 60 \div 100 = 45$

---

10. A shirt costs \$200. It is 50% off. How many dollars is the discount?

50% of 200

→ *Discount* =  $50\% \text{ of } \$200 = 50/100 \times 200 = \$100$ .

**Answer:**  $50\% \times 200 \div 100 = 100$

---

11. A store is offering a 40% discount on an item priced at \$51. How many dollars is the discount?

40% of 51

→ *Discount* =  $40\% \times \$51$ .

→ *Convert percent:*  $40/100 = \text{decimal}$ .

→ *Discount* =  $40 \times 51 / 100 = \$20$ .

**Answer:**  $40\% \times 51 \div 100 = 20$

---

12. A fundraiser has raised 46% of its \$197 goal. How many dollars have been raised?

46% of 197

→ *Amount raised* =  $46\% \times \$197$ .

→ =  $46 \times 197 / 100 = \$90$ .

**Answer:**  $46\% \times 197 \div 100 = 90$

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## Solving proportions

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13. Solve the proportion:  $\frac{1}{8} = \frac{1}{x}$ . Find  $x$ .

$$\frac{1}{8} = \frac{1}{x}$$

→ Cross-multiply:  $1 \times x = 8 \times 1 = 8$ .

→ Divide:  $x = 8 / 1 = 8$ .

**Answer:**  $x = 8 \div 1 = 8$

---

14. A car travels 12 miles on 2 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion  $\frac{2}{12} = \frac{5}{x}$  and solve.

$$\frac{2}{12} = \frac{5}{x}$$

→  $x = (12 \times 5) / 2 = 60 / 2 = 30$  miles.

**Answer:**  $x = 60 \div 2 = 30$

---

15. If 6 shirts cost \$20, how much do 6 shirts cost? Set up and solve a proportion.

$$\frac{6}{20} = \frac{6}{x}$$

→ Set up the proportion:  $6/20 = 6/x$ .

→ Cross-multiply:  $6x = 120$ .

→ Divide both sides by 6:  $x = 120/6 = 20$ .

**Answer:**  $x = 120 \div 6 = 20$

---

16. Solve the proportion:  $\frac{1}{9} = \frac{6}{x}$ . Find  $x$ .

$$\frac{1}{9} = \frac{6}{x}$$

→ Cross-multiply:  $1 \times x = 9 \times 6 = 54$ .

→ Divide:  $x = 54 / 1 = 54$ .

**Answer:**  $x = 54 \div 1 = 54$

---

17. A car travels 21 miles on 1 gallons of gas. How many miles can it travel on 6 gallons? Set up a proportion  $\frac{1}{21} = \frac{6}{x}$  and solve.

$$\frac{1}{21} = \frac{6}{x}$$

→  $x = (21 \times 6) / 1 = 126 / 1 = 126$  miles.

**Answer:**  $x = 126 \div 1 = 126$

---

18. If 6 shirts cost \$16, how much do 7 shirts cost? Set up and solve a proportion.

$$\frac{6}{16} = \frac{7}{x}$$

→ Set up the proportion:  $6/16 = 7/x$ .

→ Cross-multiply:  $6x = 112$ .

→ Divide both sides by 6:  $x = 112/6 = 56/3$ .

**Answer:**  $x = 112 \div 6 = 56/3$

---



19. Solve the proportion:  $\frac{1}{6} = \frac{4}{x}$ . Find  $x$ .

$$\frac{1}{6} = \frac{4}{x}$$

→ Cross-multiply:  $1 \times x = 6 \times 4 = 24$ .

→ Divide:  $x = 24 \div 1 = 24$ .

**Answer:**  $x = 24 \div 1 = 24$

---

20. A car travels 13 miles on 1 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion  $\frac{1}{13} = \frac{5}{x}$  and solve.

$$\frac{1}{13} = \frac{5}{x}$$

→  $x = (13 \times 5) \div 1 = 65 \div 1 = 65$  miles.

**Answer:**  $x = 65 \div 1 = 65$

---

21. If 5 shirts cost \$40, how much do 12 shirts cost? Set up and solve a proportion.

$$\frac{5}{40} = \frac{12}{x}$$

→ Set up the proportion:  $5/40 = 12/x$ .

→ Cross-multiply:  $5x = 480$ .

→ Divide both sides by 5:  $x = 480/5 = 96$ .

**Answer:**  $x = 480 \div 5 = 96$

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## Writing and simplifying ratios

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22. Write the ratio 10 to 35 in simplest form.

10 : 35

→ Write as a ratio: 10:35.

→ Find the GCF of 10 and 35: GCF = 5.

→ Divide both parts by 5: 2:7.

**Answer:**  $\div 5 \Rightarrow 2 : 7$

---

23. A bag contains 12 red marbles and 12 blue marbles. Write the ratio of red to blue in simplest form.

12 : 12

→ Ratio of red to blue: 12:12.

→ GCF of 12 and 12 = 3.

→ Simplest form: 4:4.

**Answer:**  $\div 3 \Rightarrow 4 : 4$

---

24. On a team, 12 players are seniors and 28 players are juniors. Write the ratio of seniors to juniors in simplest form.

12 : 28

→ Write the ratio of seniors to juniors: 12:28.

→ Find the GCF of 12 and 28: GCF = 4.

→ Divide both parts by 4: 3:7.

**Answer:**  $\div 4 \Rightarrow 3 : 7$

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25. Write the ratio 20 to 15 in simplest form.

20 : 15

→ Write as a ratio: 20:15.

→ Find the GCF of 20 and 15: GCF = 5.

→ Divide both parts by 5: 4:3.

**Answer:**  $\div 5 \Rightarrow 4 : 3$

---

26. A bag contains 12 red marbles and 16 blue marbles. Write the ratio of red to blue in simplest form.

12 : 16

→ Ratio of red to blue: 12:16.

→ GCF of 12 and 16 = 4.

→ Simplest form: 3:4.

**Answer:**  $\div 4 \Rightarrow 3 : 4$

---



27. On a team, 20 players are seniors and 16 players are juniors. Write the ratio of seniors to juniors in simplest form.

20 : 16

→ Write the ratio of seniors to juniors: 20:16.

→ Find the GCF of 20 and 16:  $GCF = 4$ .

→ Divide both parts by 4: 5:4.

**Answer:**  $\div 4 \Rightarrow 5 : 4$

---

28. Write the ratio 3 to 9 in simplest form.

3 : 9

→ Write as a ratio: 3:9.

→ Find the GCF of 3 and 9:  $GCF = 3$ .

→ Divide both parts by 3: 1:3.

**Answer:**  $\div 3 \Rightarrow 1 : 3$

---

29. A bag contains 4 red marbles and 16 blue marbles. Write the ratio of red to blue in simplest form.

4 : 16

→ Ratio of red to blue: 4:16.

→ GCF of 4 and 16 = 4.

→ Simplest form: 1:4.

**Answer:**  $\div 4 \Rightarrow 1 : 4$

---

30. On a team, 16 players are seniors and 28 players are juniors. Write the ratio of seniors to juniors in simplest form.

16 : 28

→ Write the ratio of seniors to juniors: 16:28.

→ Find the GCF of 16 and 28:  $GCF = 4$ .

→ Divide both parts by 4: 4:7.

**Answer:**  $\div 4 \Rightarrow 4 : 7$

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